Day-3 Configuring basic Networking hardware theory

Task-1 Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

Steps

1. Connect your laptop to your home Wi-Fi.
2. Open **Command Prompt**.
3. Run: ipconfig
4. Find **Default Gateway** under your Wi-Fi adapter. It may look like: Default Gateway . . . . . : 192.168.1.1
5. Open a web browser and enter the gateway address, for example: http://192.168.1.1
6. The router's login page should appear.
7. Enter the router's administrator username/password.
8. **Take a screenshot of the login page** and attach it to your practical file.
9. Router IP / Default Gateway: 192.168.1.1

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Task-2 Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Open **Command Prompt** and run: ipconfig

Look under the **Wireless LAN adapter Wi-Fi** section.

Example:

IPv4 Address . . . . . : 192.168.1.23

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . : 192.168.1.1

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Task-3 Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Suppose your router is:

192.168.1.1

You could temporarily configure your laptop as:

- **IP address:** 192.168.1.50
- **Subnet mask:** 255.255.255.0
- **Default gateway:** 192.168.1.1
- **Preferred DNS:** 8.8.8.8

Windows Steps

1. Open **Settings → Network & Internet → Wi-Fi**.
2. Open your connected Wi-Fi network's **Properties**.
3. Find **IP assignment**.
4. Click **Edit**.
5. Change **Automatic (DHCP)** to **Manual**.
6. Enable **IPv4**.
7. Enter the values above, **only if they match your router's network**.
8. Save the settings.
9. Open a browser and test Internet access.
10. You can also test with:

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ping 8.8.8.8

tatic IP configured: 192.168.1.50
Internet access: Working / Not Working
Ping test: Successful / Failed

Change IP assignment back to → Automatic (DHCP).

Task-4 In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Finding the DHCP Range

1. Log in to your router's admin panel.
2. Find a section such as:
    - **Network**
    - **LAN Settings**
    - **DHCP Server**
    - **DHCP Settings**
3. Note the **DHCP Start IP** and **DHCP End IP**.

Example

DHCP Start IP: 192.168.1.100

DHCP End IP: 192.168.1.200

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What happens with duplicate IP addresses?

If two devices use the **same IP address**, the network can experience an IP address conflict. The devices may lose connectivity or behave unpredictably because network traffic can be sent to the wrong device.

Two devices → Same IP → IP conflict → Connection problems